ED-01 HACKATHON • TRACK 07

PATHWISE — AI CAREER NAVIGATOR

"With Great Guidance Comes Great Opportunity!"
Mapping 1st & 2nd Year B.Tech CSE Curriculum to Industry Roles.

Project Team: Lakshya Shekhawat, Dharmesh Choudhary, Parag Sharma
Core Focus: C, DSA, DBMS, OS, & DAA Curriculum Integration

Live Web App ↗ | Share Portal ↗ | GitHub Repo ↗ | Google Form ↗ | LinkedIn ↗

01. EXECUTIVE BRIEFING

Swinging Through Career Ambiguity

Pathwise provides an intelligent career HUD for engineering students, transforming traditional coursework into practical employment readiness.

Choice Paralysis Elimination

Students wander through unstructured tutorials. Pathwise curates verified milestones based on target engineering roles.

Spider-Sense HUD Analytics

Real-time skill gap computation comparing student competencies directly against industry baseline standards.

SFIA & O*NET Taxonomies

Standardized skills mapping ensuring alignment with international occupational competency frameworks.

Adaptive Pacing Engine

Dynamic timeline recalibration tailored precisely to individual weekly available study hours (5 to 30 hrs/wk).

02. THE CHALLENGE

Core Industry Gap for CSE Freshmen

First and second-year B.Tech students face major disconnects between theoretical academia and software engineering expectations.

82%

of Engineering Freshmen report feeling unguided on how to translate university coursework into professional software development offers.

Academic Isolation

Core subjects are taught in silos without demonstrating real-world software system relevance.

Static Course Timelines

Standard curricula ignore individual student bandwidth and internship preparation deadlines.

03. SOLUTION ARCHITECTURE

How Pathwise Powers Up Students

A lightweight, browser-native Single Page Application built for speed, responsiveness, and zero server latency.

1. Interactive Skill Audit Engine

Evaluates student proficiency across 1st & 2nd-year CSE subjects and calculates immediate role readiness percentages.

2. Dynamic Roadmap Recalibration

Adjusts weekly study milestones automatically based on user input bandwidth, ensuring steady progress without burnout.

3. Integrated Feedback & Evaluation Hub

Connects seamlessly with candidate intake forms and response analytics for cohort tracking and mentorship assessment.

04. CURRICULUM MAPPING (YEAR 1)

Foundational CSE Integration

Mapping first-year university fundamentals directly to entry-level engineering capabilities.

Academic Subject	Core Topics	Industry Capability Mapping
C / C++ Programming	Pointers, Memory Allocation, Recursion	Low-level resource management & algorithmic logic
Discrete Mathematics	Propositional Logic, Graph Theory	Analytical reasoning & foundational AI logic
Engineering Physics & Math	Linear Algebra, Calculus, Matrices	Numerical computations & graphics pipelines

05. CURRICULUM MAPPING (YEAR 2)

Core Computer Science & Systems

Mapping second-year systems architecture to professional backend and algorithm engineering roles.

Academic Subject	Core Topics	Industry Capability Mapping
Data Structures & Algorithms (DSA)	Trees, Graphs, Sorting, Dynamic Programming	Optimal problem solving & technical interview readiness
Database Management Systems (DBMS)	Normalization, ACID, SQL Indexing, B-Trees	Relational data modeling & query performance tuning
Operating Systems (OS)	Process Scheduling, Concurrency, Paging	System performance, threads, & memory management
Design & Analysis of Algorithms (DAA)	Greedy algorithms, Divide and Conquer, NP-Completeness	Scalable system architecture & complexity analysis

06. TARGET DESTINATIONS

Career Tracks Accelerated

Pathwise aligns core subjects directly with three high-demand industry engineering career paths.

Software Dev (SDE-1)

Focuses on advanced DSA, system design logic, and clean code principles.

Backend Engineer

Focuses on relational databases, API development, concurrency, and caching.

AI / ML Engineer

Focuses on Python pipelines, linear algebra, and data structures.

3.5x Faster

Transition from core university coursework to internship readiness.

07. PROOF OF WORK & SHARE PORTAL

Live Web Application & Share Links

Access the fully functional live application and the shared workspace platform.

Live Web Application

Experience the interactive Pathfinder portal live in your browser.

Launch Web App ↗

Shared Platform Link

Access collaborative project workspace and discussion portal.

Open Share Portal ↗

08. CANDIDATE INTAKE

Feedback & Evaluation Hub

Integrated survey collection for student onboarding, evaluation, and mentor feedback.

Candidate Intake Google Form

Judges and mentors can submit feedback and test candidate onboarding workflows directly.

Open Google Form ↗

09. EXECUTION ROADMAP

3-Step Methodology to Success

A simple, structured workflow designed for busy undergraduate engineering students.

1. Skill Audit

Self-assess completed university coursework and mastery of core subjects.

2. Gap Analysis

Compare profile instantly against target role benchmarks and SFIA standards.

3. Execution

Follow weekly milestones and project capstones fitted to study availability.

10. RESOURCE MATRIX

Complete Project Links Summary

All verified project deliverables and creator profiles in one place.

Deliverable Asset	Resource Description	Active Link
Live Web Application	Hosted Pathfinder Portal	Open App ↗
Shared Portal Link	Collaborative Platform	Open Portal ↗
GitHub Repository	Source Code & Docs	View Repo ↗
Google Form Survey	Candidate Intake & Feedback	Open Form ↗
Team Profile	Lakshya Shekhawat, Dharmesh Choudhary, Parag Sharma	Connect ↗

READY TO TAKE THE LEAP?

THANK YOU!

Small steps today. A stronger tech career tomorrow. Explore the live application and share portal below.

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Hackathon Submission: Pathwise — AI Career Navigator

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